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Bridging Technology and Heritage: A Digital Platform for Creole Language Revitalization

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Motivation

1. Creole languages remain largely absent in Computational Linguistics (CL) and the Digital world
 2. Combining CL and Digital Humanities (DH) for Creole promotion and revitalization
 3. Make Creole Linguistic data accessible
- Proposal:** Creation of a Creole Hub to connect linguistic tools, data, and people



Context

- Creole languages are low-resource and under-represented
- Data scarcity caused by limited/restricted written and digital presence
- Language/AI models are optimized for high-resource contexts
- Building infrastructure becomes both a technical and ethical intervention



Data-Driven NLP and Structural Inequity

- Modern NLP/AI = data-driven algorithms
- Data-driven = data-hungry $\neq$ low-resource
- Each iteration of AI models based on increased parameters/more data.
 - Deepens pre-existing inequalities
- Since massive-scale data collection is impossible for Creoles, our path forward cannot be quantitative
 - Going back to Linguistics for qualitative supervised modeling
 - Going back to communities for typologically and culturally accurate linguistic modeling



Where Computational Models Fail

- My initial Creole NLP experiments with pre-existing multilingual models exposed systematic errors.
 - Tagging confusion on polyfunctional items, which are ubiquitous in French-based Creole.
 - Parsing failures on clause boundaries in Creole due to low morphological cues, a property of Creoles.
- Current architectures learn surface form without linguistic function



Healthcare Findings: Why Community Matters

- Healthcare AI translation experiments for Haitian Creole confirmed that grammaticality alone is insufficient. (Mompelat 2025c)
 - Public-health texts collapse register/variation and use terminology unfamiliar to target audiences.
 - Varieties of HC reveal socio-linguistic and cultural stratification that models ignore: choosing one variety could alienate some patients.
- “putting linguistics back in” is necessary but must be paired with sociolinguistics and community efforts.

Steps to improving current NLP models

- **Data annotation:** POS, dependency, and translation-quality labeling.
 - must reflect sociolinguistic realities and community feedback
- **Data augmentation:** expands coverage and improves predictions.
 - must be based on the typological challenges of the language and can/should be supported by LLMs provided a Human-in-the-Loop (HIL) approach
- **Result:** More data, more effectively collected/generated/preprocessed, more domain/variety-specific, more community/culturally-grounded
→ Smarter and more ethical language models and AI solutions for Creoles.
- Refinement remains iterative: corpus → model → analysis → improved corpus



Why a Creole Hub

- Individual tools and datasets are valuable, but isolated.
- The Creole Hub centralizes corpora, models, tools, and annotation in one public interface.
- Enables collaboration among researchers, educators, and speakers.
- In under-funded contexts, collaboration is not optional but a survival strategy.
- Infrastructure as scholarship: an ethical, participatory design for linguistic justice. (Rockwell 2011)

Dataset Hosting

- Datasets will be stored on a server and a Github Repo and made available on the hub for ease of access
 - Parallel Corpora with metadata for sources

Carnegie Mellon University Haitian Creole language data :

- Original (CMU 2010): 2 files, 1 HC and 1 ENG, 1 sentence per line
- Processed and annotated version (Mompelat et al. 2025): 1 csv file

1	en	a bit everywhere
1	ht	enpe tout kote
2	en	a bit here
2	ht	enpe la
3	en	about a couple hours ago it was about a nine
3	ht	apeprè kèk èdtan de sa li te apeprè nevè

– Webscraped available datasets

Leipzig Corpora Collection (2017): *Haitian Creole Community corpus based on material crawled in 2017*. Leipzig Corpora Collection. Dataset. https://corpora.uni-leipzig.de/en?corpusId=hat_community_2017.

Dataset Hosting

- Mompelat (2021)
- Sociolinguistic interviews (the Frog story)

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2^{ème} intervenant (fille)

Alow, man ka wè an tibolonm, épi i ka gadé an..<an sé> an ti-krapolad<An krapolad oben an grenouy kriyé'y kon'w lé-a> épi an ti-chien ki anlè'y, épi an ti-po, épi tibolonm lan i chanm-li, épi sé lannuit <Kisa ka fè di sa, han han ki détay ka fèw> épi kwasan lalin épi i ni an bel chanm kanmenm, pas i ni an bel kabann, kabann lan, kon yo ka kriyé sa, an li a baldaken <han han toutafè> épi i ni an bel ti-luminèw, an bel ti-limiè ka tonbé anlè'y, donk man ka pansé sé an ti-manmay ki dwet adan an famiy, ki ni lajan<ézé wè !> épi, épi kisa man pé di dot, man pa ka wè atè-a ni an bagay, mé man pa ka wè kisa kila <lenj-li ki la, nou pé di sa sé bot-li, soulié'y> soulié'y ki la <Ok, é la ?>

Universal Dependencies

- Universal Dependencies corpus
- https://github.com/UniversalDependencies/UD_Haitian_Creole-Adolphe

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```
# sent_id = 3225
# text = Si w vin fè sa ki byen, èske w pap gen favè m?
1  Si  si  SCONJ  _  _  4  mark  _  _
2  w  ou  PRON  _  Number=Sing|Person=2|PronType=Prs  4  nsubj  _  _
3  vin  vini  VERB  _  _  4  compound:svc  _  _
4  fè  fè  VERB  _  _  12  advcl  _  _
5  sa  sa  PRON  _  _  4  obj  _  _
6  ki  ki  PRON  _  PronType=Rel  7  nsubj  _  _
7  byen  byen  ADV  _  _  5  acl:relcl  _  SpaceAfter=No
8  ,  ,  PUNCT  _  _  4  punct  _  _
9  èske  èske  ADV  _  _  12  advmod  _  _
10 w  ou  PRON  _  Number=Sing|Person=2|PronType=Prs  12  nsubj  _  _
11 pap  pa  ADV  _  Polarity=Neg  12  advmod  _  _
12 gen  gen  VERB  _  _  0  root  _  _
13 favè  favè  NOUN  _  _  12  obj  _  _
14 m  mwen  PRON  _  Number=Sing|Person=1|PronType=Prs  13  det  _  SpaceAfter=No
15 ?  ?  PUNCT  _  _  12  punct  _  _
```

Core NLP Tools Integrated in the Hub

- Corpus Linguistic Tools :
 - Corpus exploration and querying (metadata, linguistic search, REGEX)
- Foundational modular toolkit:
 - **Language ID** – detects Creole/French/Other tokens. (Mompelat 2025a)
 - **POS Tagger** – tokenizes and labels grammatical roles. (Mompelat 2025b)
 - **Dependency Parser** – maps syntactic relations (UD). (Mompelat 2025d)
 - **Machine Translation** – translates Creole ↔ French/English with confidence scores (Mompelat 2025e)
- NLP Toolkit (lemmatizer, morphological analyzer, name entity)
- All components share a JSON schema for consistent input / output.



Example Pipeline and JSON Output



```
{
  "text": "Sa sé on bel kay.",
  "metadata": {
    "source_lang": "gcf",
    "target_lang": "fra",
    "models": {
      "lid": "CreoleLID_v1.0",
      "pos": "CreolePOS_v1.0",
      "parser": "CreoleParser_v1.0",
      "mt": "CreoleMT_v1.0"
    }
  },
```

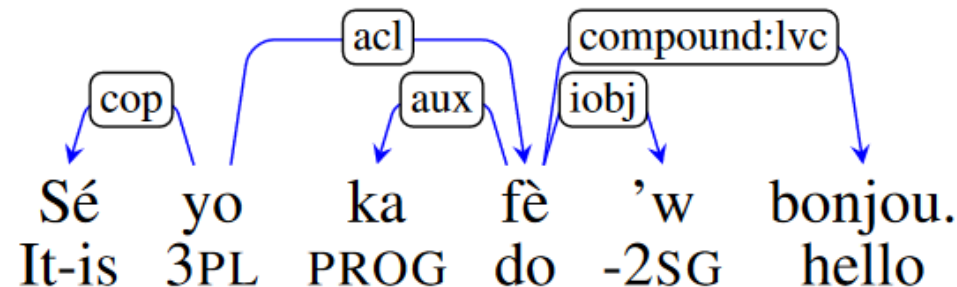
```
"pipeline": {
  "lid": {
    "language": "gcf",
    "confidence": 0.97,
    "per_token": [
      {"token": "Sa", "lang": "gcf", "conf": 0.98},
      {"token": "sé", "lang": "gcf", "conf": 0.96},
      {"token": "on", "lang": "gcf", "conf": 0.97},
      {"token": "bel", "lang": "gcf", "conf": 0.95},
      {"token": "kay", "lang": "gcf", "conf": 0.99},
      {"token": ".", "lang": "gcf", "conf": 0.99}
    ]
  },
  "mt": {
    "source": "Sa sé on bel kay.",
    "target": "C'est une belle maison.",
    "confidence": 0.94
  }
}
```



Example Pipeline and JSON Output

```
"pos": {
  "tokens": ["Sé", "yo", "ka", "fè", "'w", "bonjou", "."],
  "pos_tags": ["AUX", "PRON", "PART", "VERB", "PRON", "NOUN", "PUNCT"],
  "lemmas": ["sé", "yo", "ka", "fè", "ou", "bonjou", "."]
},
"parser": {
  "dependencies": [
    {"id": 1, "form": "Sé", "lemma": "sé", "upos": "AUX", "head": 2, "dep": "cop"},
    {"id": 2, "form": "yo", "lemma": "yo", "upos": "PRON", "head": 0, "dep": "root"},
    {"id": 3, "form": "ka", "lemma": "ka", "upos": "AUX", "head": 4, "dep": "aux"},
    {"id": 4, "form": "fè", "lemma": "fè", "upos": "VERB", "head": 2, "dep": "acl"},
    {"id": 5, "form": "'w", "lemma": "ou", "upos": "PRON", "head": 4, "dep": "iobj"},
    {"id": 6, "form": "bonjou", "lemma": "bonjou", "upos": "NOUN", "head": 4, "dep": "compound:lvc"},
    {"id": 7, "form": ".", "lemma": ".", "upos": "PUNCT", "head": 1, "dep": "punct"}
  ]
}
```

Token	POS
Sé	AUX
yo	PRON
ka	PART
fè	VERB
'	PUNCT
w	PRON
bonjou	NOUN
.	PUNCT





Creole Hub

Phase 1 : Access and Propagation	Phase 2 : Collaboration and Ownership	Phase 3 : Community and Crowdsourcing
- Publicly accessible prototype - Input text, choose tools, get JSON outputs - Read-only datasets, metadata search	- User management and research functions - Corpora distribution, analysis running, result storing - Metadata and credit tracking - From showcase to collaborative infrastructure	- Participation from educators, institutions and speakers - Annotation and validation crowdsourcing - Feedback loops from local expertise to model improvement - Digital participation as linguistic empowerment

Hub 1.0: by Spring 2026

Hub 2.0: by Spring 2027

Hub 3.0: by Spring 2028

Technical Architecture: Building for Longevity

- **Backend:** FastAPI (Python) with Pydantic schema validation
- **Frontend:** React interface (modular, API-driven)
- **Data storage:**
 - Open datasets hosted on **GitHub** (light, reproducible, version-controlled)
 - Secure corpora + models on **University HPC** (restricted, high-performance)
- **Dockerization** for portability and scaling.
- **Architecture:** Modular, CMS-free design → API | Frontend | Data layers communicate through JSON
- Transparency and openness translate humanistic values into code



Conclusion: From Tools to Humanistic Computation

- Computation should extend, not replace, linguistic understanding.
- Re-centers linguistics within NLP through typology, variation, and collaboration.
- Combining computational methods with humanistic design
- For DH, views data work as cultural stewardship
- For CL, it models context-aware, interpretable, and equitable systems.

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Thank you 😊

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Want to read more and/or
collaborate?

Find me on LinkedIn!



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